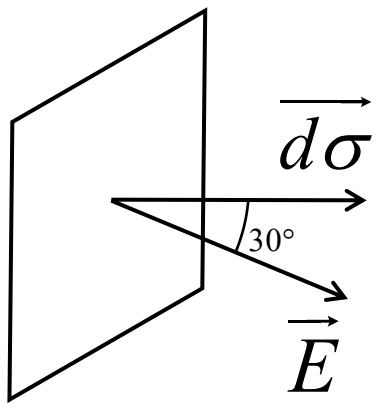
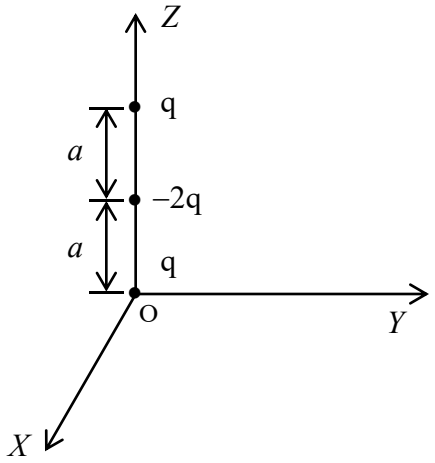
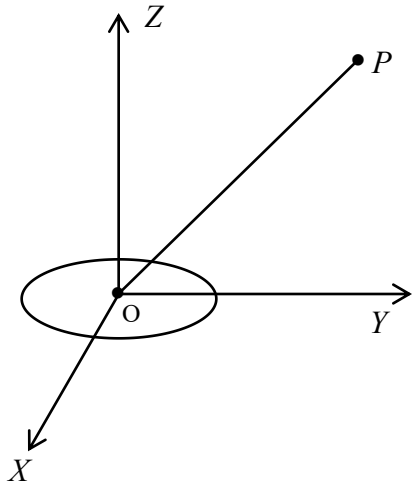
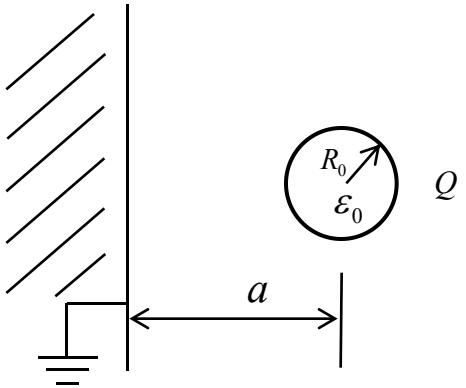
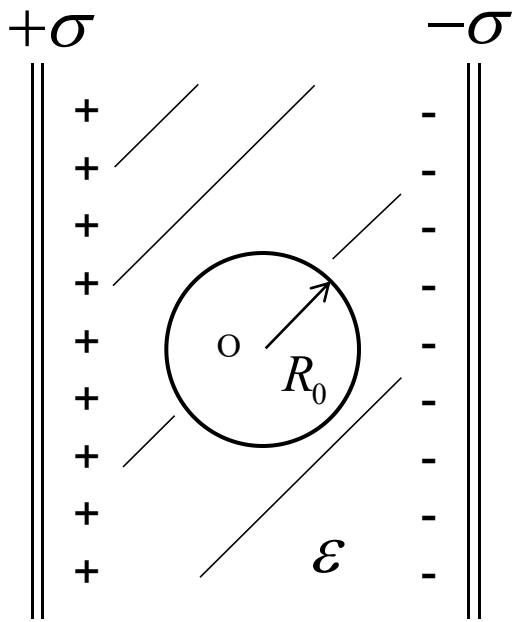
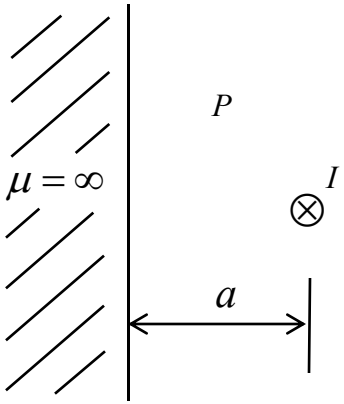

$\overrightarrow{d\sigma}$
30°
$\vec{E}$

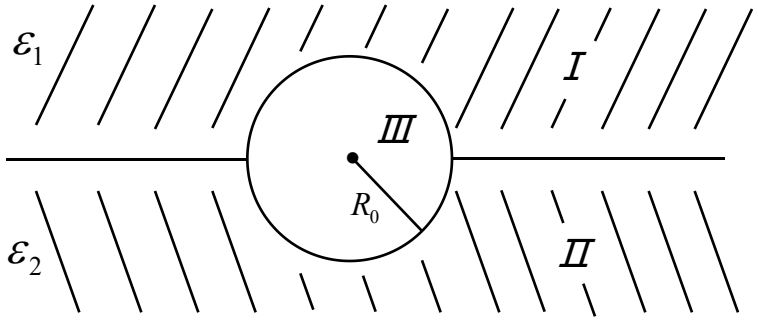

Z
q
a
−2q
a
q
O
Y
X

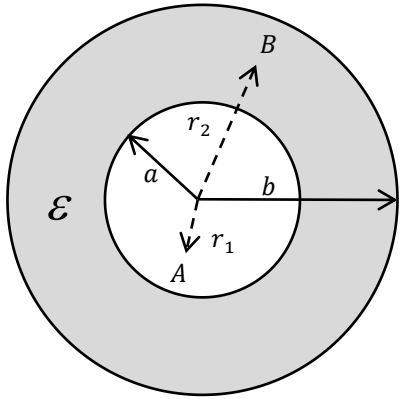

Z
P
O
Y
X

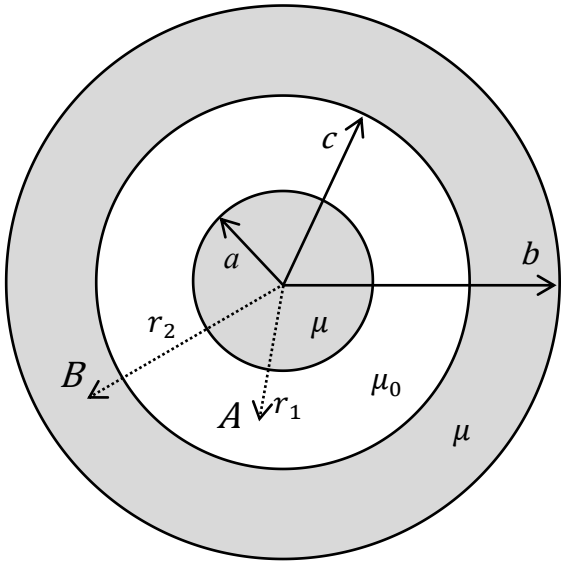

R_0
ε_0
Q
a

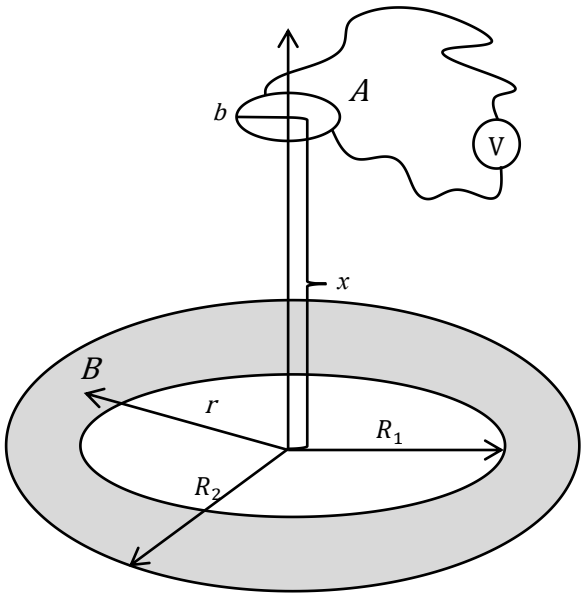

$+\sigma$
$-\sigma$
O
R_0
ε

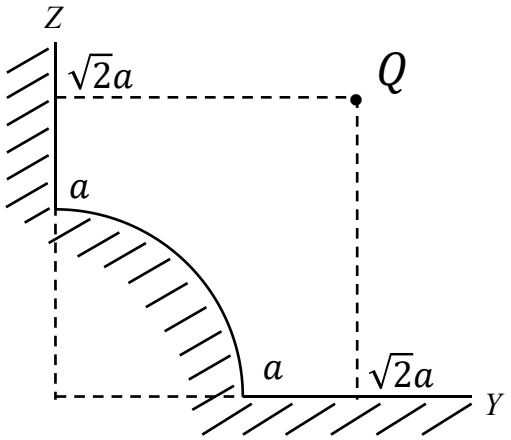

P
$\mu = \infty$
I
a

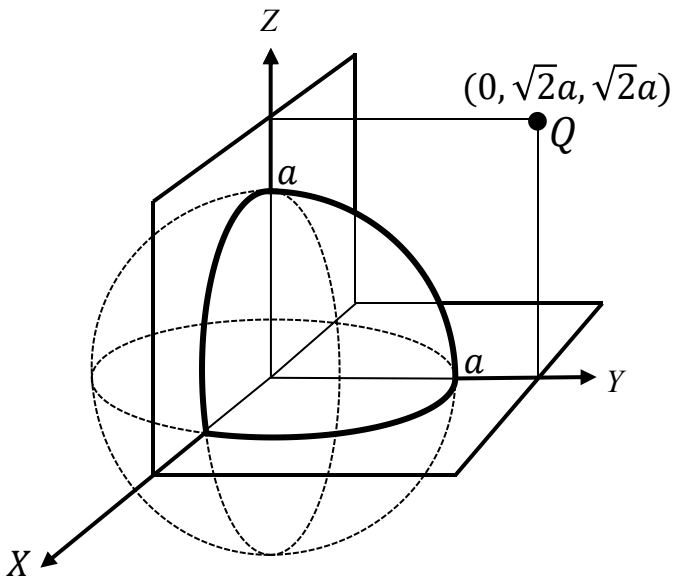

ε_1
I
III
R_0
ε_2
II

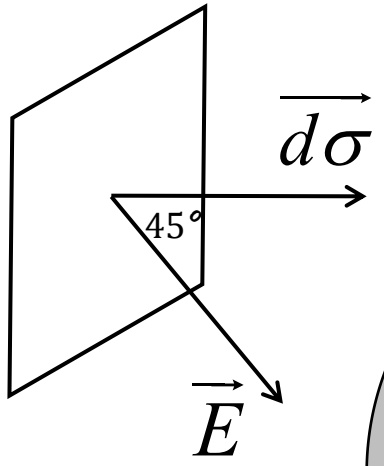

B
r_2
a
b
ε
r_1
A

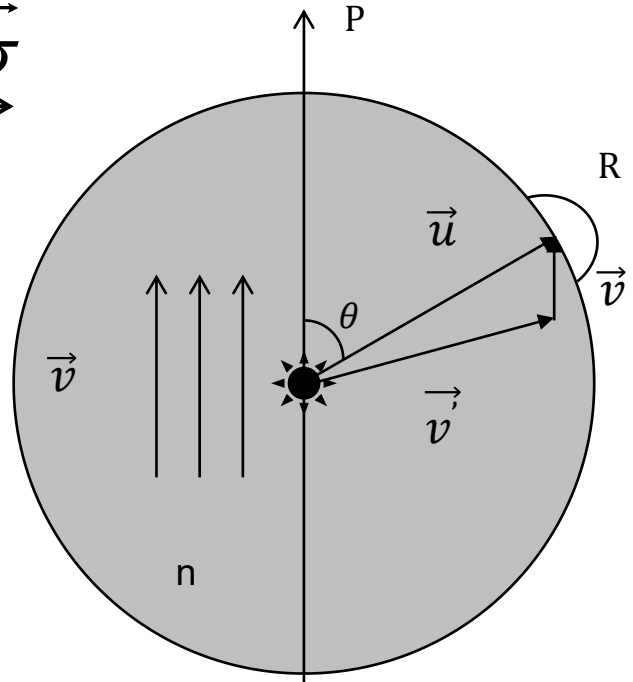

c
a
b
r_2
μ
B
μ_0
A
r_1
μ


b
A
V
x
B
r
R_1
R_2

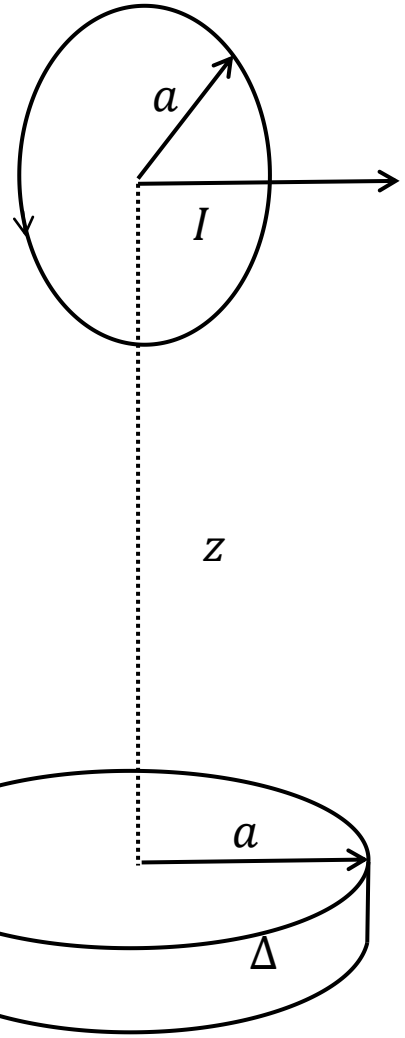

Z
$\sqrt{2}a$
Q
a
a
$\sqrt{2}a$
Y

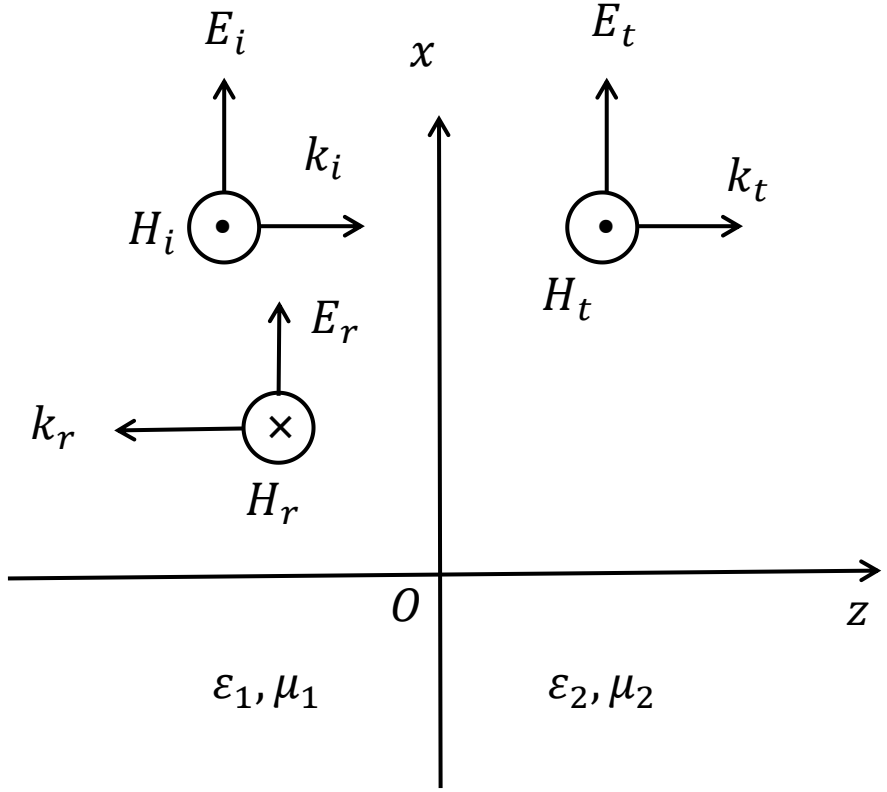

Z
$(0, \sqrt{2}a, \sqrt{2}a)$
Q
a
a
Y
X

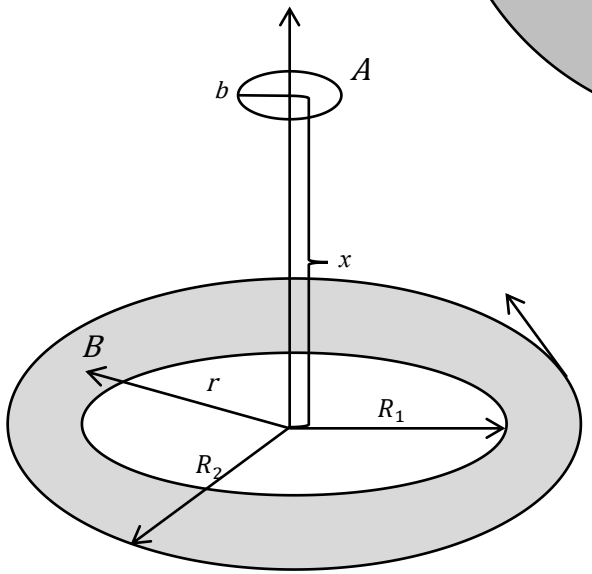

$\overrightarrow{d\sigma}$
45°
$\vec{E}$

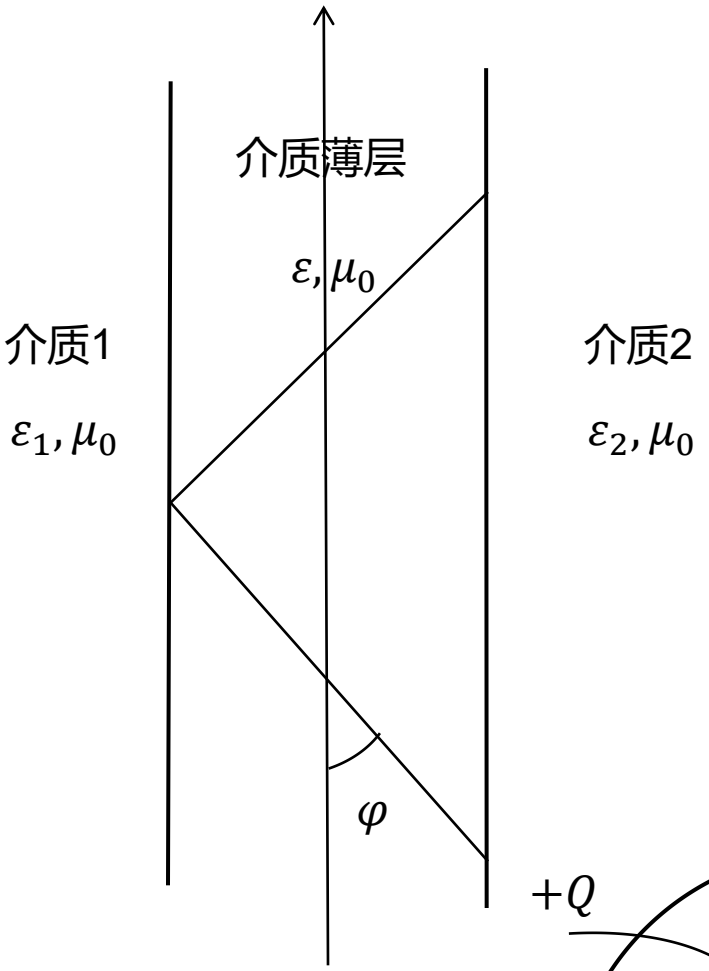

P
R
$\vec{u}$
$\vec{v}$
θ
$\vec{v}$
$\vec{v'}$
n

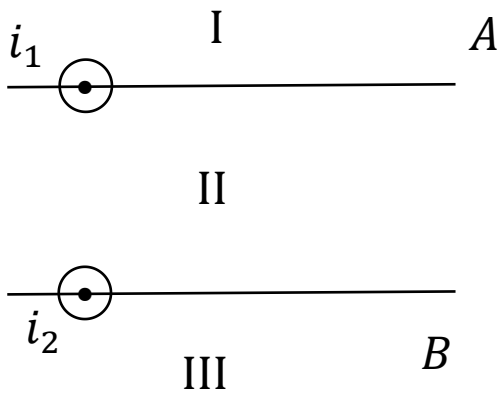
$$|v'| = \frac{c}{n} = v$$

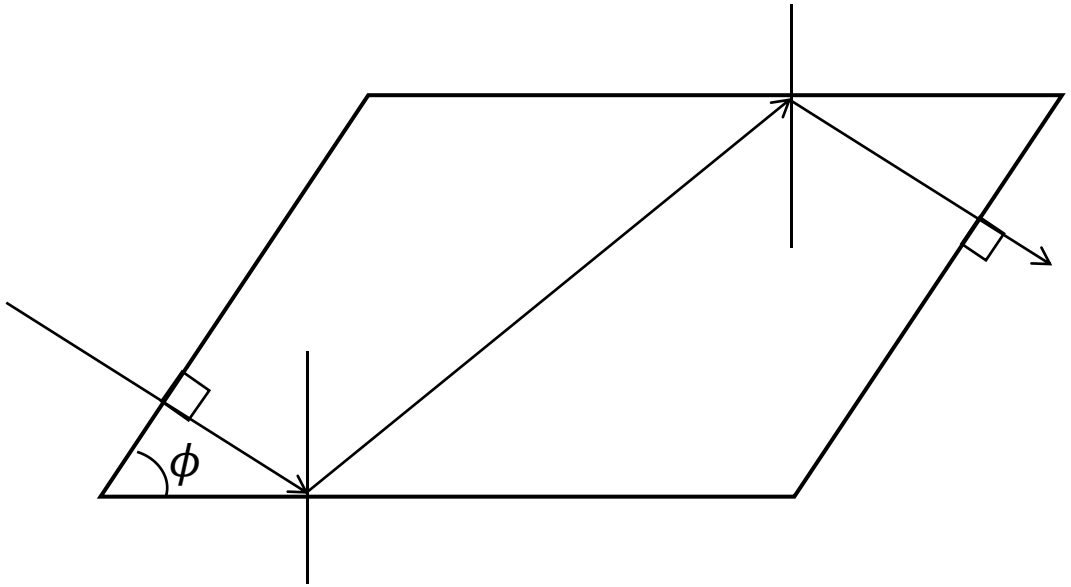

a
I
z
a
Δ

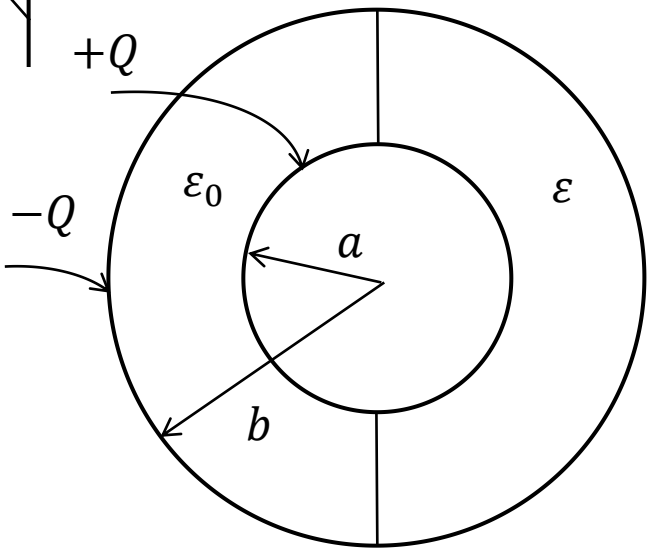

E_i
x
E_t
k_i
k_t
H_i
H_t
E_r
k_r
H_r
O
z
ε_1, μ_1
ε_2, μ_2

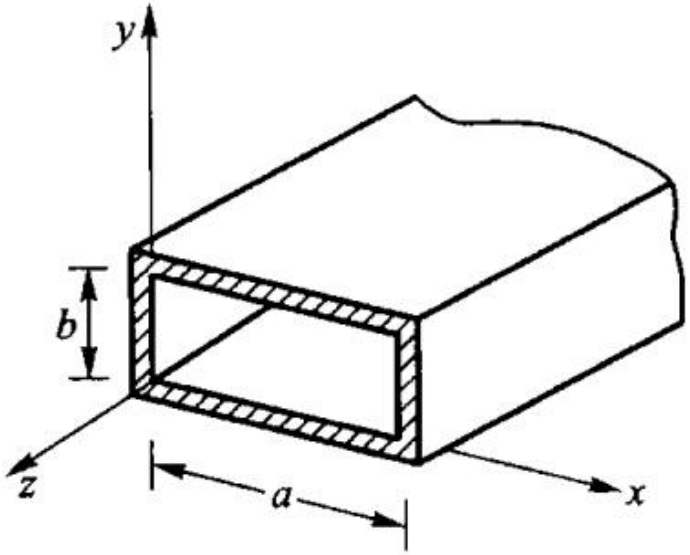

b
A
x
B
r
R_1
R_2


介质薄层
ε, μ_0
介质1
ε_1, μ_0
介质2
ε_2, μ_0
φ


i_1
I
A
II
i_2
III
B


ϕ


$+Q$
$-Q$
ε_0
ε
a
b


y
b
z
a
x